

SELinux Basics Theory (1 hour):

Introduction to SELinux and its role in server security. Understanding SELinux modes and contexts.

Practical (1 hour): Enable SELinux and troubleshoot access denials.

Task: Document an SELinux configuration with resolved issues.

TASKS

1. Check the current SELinux status on your Linux server using the `getenforce` and `sestatus` commands, and write down which mode (Enforcing, Permissive, or Disabled) is currently active.

Check SELinux Status

Step 1: Check Current Mode

`getenforce`

Step 2: Detailed Status

`sestatus`

```
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:       /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:         enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     33
[root@localhost ~]#
```

**2. Switch SELinux to Enforcing mode using `setenforce` and verify the change with `getenforce`.
Hint: You may need `sudo` privileges to change the mode.**

Switch to Enforcing Mode

`setenforce 1`

Verify:

getenforce

```
[root@localhost ~]# setenforce 1
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]#
```

3. Simulate an SELinux access denial by creating a new directory (e.g., /tmp/testapp), setting its context to a type not allowed for your web server (such as user_home_t), and then try to serve a file from it using your web server. Note the error message in /var/log/audit/audit.log or /var/log/messages.

Simulate an SELinux Access Denial

Step 1: Create a Test Directory

```
mkdir -p /tmp/testapp
```

Step 2: Create an HTML File

```
echo "<h1>SELinux Test</h1>" > /tmp/testapp/index.html
```

Step 3: Give the Directory the Wrong SELinux Context

```
chcon -t user_home_t /tmp/testapp
```

Check the context:

```
ls -Zd /tmp/testapp
```

```
[root@localhost ~]# ls -Zd /tmp/testapp
unconfined_u:object_r:user_home_t:s0 /tmp/testapp
[root@localhost ~]#
```

Step 4: Copy the File to Apache Web Root

```
cp /tmp/testapp/index.html /var/www/html/
```

Step 5: Give the File the Wrong Context

```
chcon -t user_home_t /var/www/html/index.html
```

Verify:

```
ls -Z /var/www/html/index.html
```

```
[root@localhost ~]# ls -Z /var/www/html/index.html
unconfined_u:object_r:user_home_t:s0 /var/www/html/index.html
[root@localhost ~]#
```

Step 6:

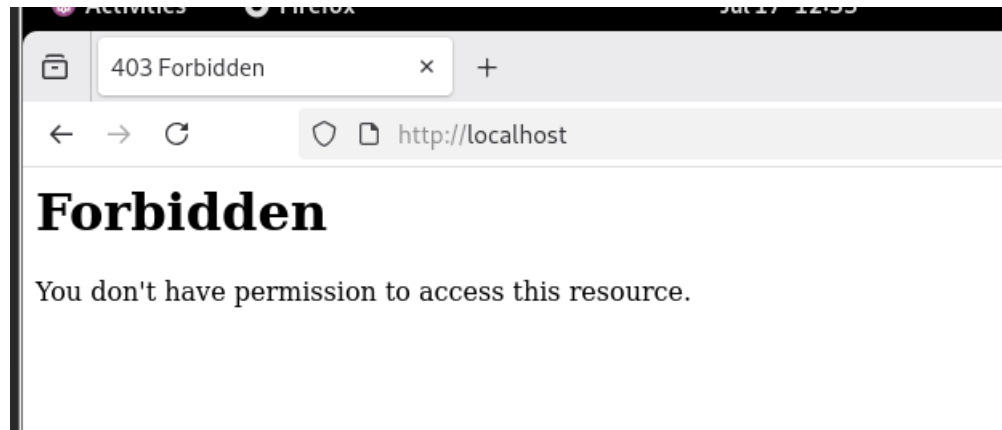
Open the Browser

http://localhost

Step 7: Check the Audit Log

tail -20

/var/log/audit/audit.log



4. Use the chcon

```
unconfined_u:object_r:user_home_t:s0 /var/www/html/index.html
[root@localhost ~]# tail -20 /var/log/audit/audit.log
type=CRED_REFR msg=audit(1784271459.235:805): pid=9446 uid=0 auid=0 ses=2 subj=s
system_u:system_r:xdm_t:s0-s0:c0.c1023 msg='op=PAM:setcred grantors=pam_unix,pam
gnome_keyring acct="root" exe="/usr/libexec/gdm-session-worker" hostname=localho
st.localdomain addr=? terminal=/dev/tty1 res=success'UID="root" AUID="root"
type=SERVICE_STOP msg=audit(1784271488.016:806): pid=1 uid=0 auid=4294967295 ses
=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=fprintd comm="systemd" ex
e="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root"
AUID="unset"
type=BPF msg=audit(1784271488.032:807): prog-id=80 op=UNLOAD
type=BPF msg=audit(1784271802.782:808): prog-id=81 op=LOAD
type=BPF msg=audit(1784271802.783:809): prog-id=82 op=LOAD
type=BPF msg=audit(1784271802.783:810): prog-id=83 op=LOAD
type=SERVICE_START msg=audit(1784271803.448:811): pid=1 uid=0 auid=4294967295 se
s=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=systemd-timedated comm="
systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success
'UID="root" AUID="unset"
type=SERVICE_START msg=audit(1784271804.085:812): pid=1 uid=0 auid=4294967295 se
s=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatche
r comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res
=success'UID="root" AUID="unset"
type=SERVICE_STOP msg=audit(1784271821.977:813): pid=1 uid=0 auid=4294967295 ses
```

command to assign the correct SELinux context (e.g., httpd_sys_content_t) to your test directory or file so your web server can access it, then confirm the issue is resolved by reloading the page successfully.

Fix the SELinux Context

Step 1: Change the Context

```
chcon -t httpd_sys_content_t /var/www/html/index.html
```

Step 2: Verify

`ls -Z /var/www/html/index.html`

```
[root@localhost ~]# ls -Z /var/www/html/index.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html
[root@localhost ~]#
```

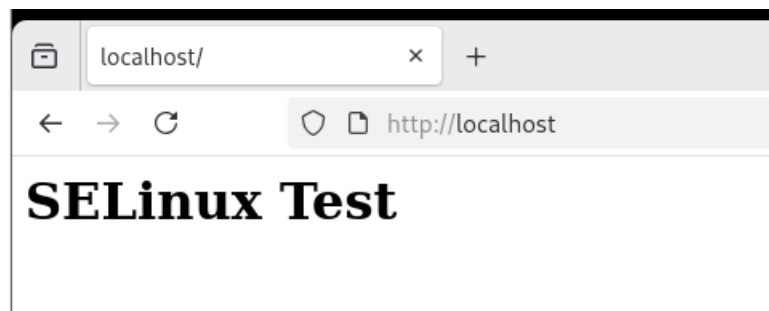
Step 3: Restart Apache

`systemctl restart httpd`

Step 4: Refresh the Browser

Open:

`http://localhost`



5. Document the steps you took to identify and resolve the SELinux access denial, including the commands used, the error log entry, and a brief explanation of how SELinux contexts control access.

Documentation

```
262 getenforce
263 sestatus
264 setenforce 1
265 getenforce
266 mkdir -p /tmp/testapp
267 echo "<h1>SELinux Test</h1>" > /tmp/testapp/index.html
268 chcon -t user_home_t /tmp/testapp
269 ls -Zd /tmp/testapp
270 cp /tmp/testapp/index.html /var/www/html/
271 chcon -t user_home_t /var/www/html/index.html
272 ls -Z /var/www/html/index.html
273 tail -20 /var/log/audit/audit.log
274 chcon -t httpd_sys_content_t /var/www/html/index.html
275 ls -Z /var/www/html/index.html
276 systemctl restart httpd
277 history
[root@localhost ~]#
```